Indian Statistical Institute

Parametric Inference

Mock Semestral Examination

B.Stat. 3rd Year · Semester 1, 2026–27

Time: 3 hours **Maximum marks:** 100

Instructions.

- Answer **all** questions. Marks are shown in brackets against each part.
- State clearly any theorem you invoke, and verify that its hypotheses hold. Quoting Lehmann–Scheffé, Basu or Karlin–Rubin without checking completeness, ancillarity or MLR respectively will not earn full credit.
- Throughout, Φ and φ denote the standard normal cdf and pdf.
- You may use, without proof, the following values:

$$z_{0.05} = 1.645, \qquad z_{0.025} = 1.960, \qquad \Phi(0.98) = 0.8365, \qquad \Phi(1.645) = 0.9500.$$

- Unless stated otherwise, "risk" means mean squared error and "loss" means squared error loss.

Q1. [**12 marks**]

(a) State the Neyman–Fisher factorisation theorem. [**3**]

(b) State and prove the Rao–Blackwell theorem. State precisely when the improvement is strict. [**6**]

(c) Give, with justification, an example of a family of distributions possessing a sufficient statistic that is *not* complete. [**3**]

Q2. [**12 marks**]

Let $X \sim \text{Bin}(m, \theta)$ with m known and $0 < \theta < 1$ unknown.

(a) Show that X is a complete sufficient statistic for θ. [**4**]

(b) For $1 \le k \le m$, find the UMVUE of θ^k. [**5**]

(c) Show that $\psi(\theta) = \theta^{m+1}$ admits no unbiased estimator. [**3**]

Q3. [**12 marks**]

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Poisson}(\lambda)$, $\lambda > 0$.

(a) Find the UMVUE of $\mathbb{P}_\lambda\{X_1 = 0\} = e^{-\lambda}$. [**6**]

(b) Compute the variance of your estimator. [**3**]

(c) Does it attain the Cramér–Rao lower bound? Justify your answer, and comment on what happens as $n \to \infty$. [**3**]

Q4. **[10 marks]**

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$ with $n \geq 3$ and both parameters unknown. Write $S^2 = \frac{1}{n-1} \sum_i (X_i - \overline{X})^2$ and $R = X_{(n)} - X_{(1)}$ for the sample range.

(a) Show that $\left(\sum_i X_i,\ \sum_i X_i^2\right)$ is complete and sufficient. **[4]**

(b) Show that $V = R/S$ is ancillary. **[3]**

(c) Deduce that $\mathbb{E}[R] = \mathbb{E}[V]\,\mathbb{E}[S]$. **[3]**

Q5. **[12 marks]**

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} f(x \mid \theta) = \frac{1}{\theta} e^{-x/\theta}$, $x > 0$, $\theta > 0$. Consider the inverse-gamma family of priors

$$\pi_{\alpha,\beta}(\theta) \;\propto\; \theta^{-(\alpha+1)} e^{-\beta/\theta}, \qquad \alpha, \beta > 0.$$

(a) Show that this family is conjugate and identify the posterior. **[5]**

(b) Find the Bayes estimate of θ under squared error loss, stating any condition needed for it to exist. **[3]**

(c) Is the Bayes estimate unbiased for θ? **[2]**

(d) Find the generalized Bayes estimate of θ under the improper prior $\pi(\theta) = 1/\theta$, and comment on how it compares with $\overline{X}$. **[2]**

Q6. **[12 marks]**

Let $X \sim \text{Bin}(n, \theta)$, $0 < \theta < 1$, and let the prior be $\text{Beta}(p, q)$.

(a) Show that the Bayes estimate of θ under squared error loss is $T_{p,q} = \dfrac{p + X}{p + q + n}$. **[3]**

(b) Compute $R(\theta, T_{p,q})$ and determine all (p, q) for which the risk is constant in θ. **[6]**

(c) Hence identify a minimax estimator of θ, and compare its maximum risk with that of the UMVUE X/n. **[3]**

Q7. **[12 marks]**

A single observation X is available. Consider

$$H_0 :\; X \sim N(0,1) \qquad \text{against} \qquad H_A :\; X \sim N(0,4).$$

(a) Derive the form of the most powerful test of level α. **[6]**

(b) For $\alpha = 0.05$, state the rejection region explicitly and compute the power. **[3]**

(c) Now let $X \sim N(0, \sigma^2)$ with $\sigma^2 > 0$ unknown. Does a UMP test of size α exist for $H_0 : \sigma^2 = 1$ against $H_A : \sigma^2 \neq 1$? Justify carefully. **[3]**

Q8. **[10 marks]**

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Uniform}[0, \theta]$, $\theta > 0$.

(a) Show that the family has monotone likelihood ratio in $X_{(n)}$. **[4]**

(b) Derive the UMP test of size α for $H_0 : \theta \leq \theta_0$ against $H_A : \theta > \theta_0$. **[4]**

(c) Write down its power function and verify that the size is attained at $\theta = \theta_0$. **[2]**

Q9. **[10 marks]**

(a) Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Cauchy}(\theta)$, i.e. with density $\frac{1}{\pi\{1+(x-\theta)^2\}}$. Show that $\overline{X}$ is *not* a consistent estimator of θ, whereas the sample median is. [**6**]

(b) In the model $Y_i = \alpha + \beta x_i + \varepsilon_i$, $i = 1, \ldots, n$, with x_i known constants and ε_i iid with mean 0 and variance σ^2, state and justify the condition on $\{x_i\}$ under which the least squares estimate $\hat{\beta}_{LS}$ is consistent for β. [**4**]

End of paper